Efficacy, Safety, and Accuracy of Intra-articular Therapies for Hand Osteoarthritis: Current Evidence.

PMID: 36633823 | Indexed in PubMed

The lifetime risk of symptomatic hand osteoarthritis (OA) is 39.8%, with one in two women and one in four men developing the disease by age 85 years and no disease-modifying drug (DMOAD) available so far. Intra-articular (IA) therapy is one of the options commonly used for symptomatic alleviation of OA disease as it can circumvent systemic exposure and potential side effects of oral medications. The current narrative review focuses on the efficacy and safety profiles of the currently available IA agents in hand OA (thumb-base OA or interphalangeal OA) such as corticosteroids and hyaluronic acid (HA), as well as the efficacy and safety of IA investigational injectates in phase 2/3 clinical trials such as prolotherapy, platelet-rich plasma, stem cells, infliximab, interferon-? and botulinum toxin, based on the published randomized controlled trials on PubMed database. The limited published literature revealed the short-term symptomatic benefits of corticosteroids in interphalangeal OA while long-term data are lacking. Most of the short-term studies showed no significant difference between corticosteroids and hyaluronic acid in thumb-base OA, usually with a faster onset of pain relief in the corticosteroid group and a slower but greater (statistically insignificant) pain improvement in the HA group. The majority of studies in investigational agents were limited by small sample size, short-term follow-up, and presence of serious side effects. In addition, we reported higher accuracy rates of drug administrations under imaging guidance than landmark guidance (blind method), and then briefly describe challenges for the long-term efficacy and prospects of IA therapeutics.

Botulinum toxin type a combined with transcranial direct current stimulation reverses the chronic pain induced by osteoarthritis in rats.

PMID: 35421437 | Indexed in PubMed

Osteoarthritis (OA) is the most common cause to lead to chronic pain. Sensitization of pain pathways including central sensitization and peripheral sensitization has been regarded as a major cause of OA pain refractory to treatment. Addressing peripheral sensitization or central sensitization alone may not adequately treat OA pain. In our previous studies, botulinum toxin type A (BoNT/A) has been shown to reduce peripheral sensitization for analgesic effects. In addition, transcranial direct current stimulation (tDCS) has also been suggested to reduce central sensitization for analgesia. The present study was designed to investigate whether BoNT/A in combination with tDCS has better analgesic effects than isolated treatment to alleviate OA-induced chronic pain in rats. The Von Frey and hot plate tests were applied to assess the pain-related behaviors at different time points. The expression level of N-methyl-D-aspartate receptor-2B (NMDAR2B) was evaluated in midbrain periaqueductal gray (PAG) by Western blot the Immunohistochemistry staining after different treatments. The results showed that the combined treatment of BoNT/A and tDCS better improved the pain-related behaviors and significantly increased the expression level of NMDAR2B protein in PAG than each isolated treatment. These results suggested that the combined treatments for relief of chronic pain were more obvious than each isolated treatment. The combination of BoNT/A and tDCS may relieve pain by increasing N-methyl-D-aspartate (NMDA) receptors in the PAG, and then the descending inhibitory systems were activated to modulate peripheral and central sensitization.

Efficacy of Intra-Articular Injection of Botulinum Toxin Type A (IncobotulinumtoxinA) in Temporomandibular Joint Osteoarthritis: A Three-Arm Controlled Trial in Rats.

PMID: 37104199 | Indexed in PubMed

Temporomandibular disorders (TMD) are complex pathologies responsible for chronic orofacial pain. Intramuscular injection of botulinum toxin A (BoNT/A) has shown effectiveness in knee and shoulder osteoarthritis, as well as in some TMDs such as masticatory myofascial pain, but its use remains controversial. This study aimed to evaluate the effect of intra-articular BoNT/A injection in an animal model of temporomandibular joint osteoarthritis. A rat model of temporomandibular osteoarthritis was used to compare the effects of intra-articular injection of BoNT/A, placebo (saline), and hyaluronic acid (HA). Efficacy was compared by pain assessment (head withdrawal test), histological analysis, and imaging performed in each group at different time points until day 30. Compared with the rats receiving placebo, those receiving intra-articular BoNT/A and HA had a significant decrease in pain at day 14. The analgesic effect of BoNT/A was evident as early as day 7, and lasted until day 21. Histological and radiographic analyses showed decrease in joint inflammation in the BoNT/A and HA groups. The osteoarthritis histological score at day 30 was significantly lower in the BoNT/A group than in the other two groups ($p = 0.016$). Intra-articular injection of BoNT/A appeared to reduce pain and inflammation in experimentally induced temporomandibular osteoarthritis in rats.

The efficacy and safety of intra-articular botulinum toxin type A injection for knee osteoarthritis: A meta-analysis of randomized controlled trials.

PMID: 36640812 | Indexed in PubMed

The purpose of this study was to investigate the efficacy and safety of intra-articular Botulinum Toxin type A (BTA) injection in the management of patients with knee osteoarthritis (KOA). The literature retrieval was conducted based on PRISMA guidelines. Databases including Pubmed, Web of science, EMBASE, and Cochrane Library were searched to identify RCTs that comparing the effects of intra-articular BTA injection with control interventions on patients with KOA. The primary outcomes involved pain and function improvements as well as the occurrence of adverse events. Seven RCTs comprising 548 participants were included in this meta-analysis. Compared with the control group, BTA injection exhibited greater pain reduction at 4 weeks posttreatment (SMD = -0.86, 95% CI [-1.52, -0.19], $p = 0.011$), but not 8-24 weeks posttreatment (wk 8, SMD = -0.53, 95% CI [-1.21, 0.15], $p = 0.127$; wk 12, SMD = -0.34, 95% CI [-0.73, 0.04], $p = 0.081$; wk 24, SMD = -0.65, 95% CI [-1.52, 0.22], $p = 0.144$). Additionally, no differences were found between BTA injection versus control intervention on functional improvement at all time points assessed (wk 4, WMD = -5.16, 95% CI [-12.31, 2.00], $p = 0.158$; wk 8, WMD = -0.98, 95% CI [-5.66, 3.71], $p = 0.683$; wk 12, WMD = -2.52, 95% CI [-7.54, 2.50], $p = 0.325$); wk 24, WMD = -3.66, 95% CI [-14.09, 6.76], $p = 0.491$). There was no significant difference in adverse event rate between the BTA and control group (OR = 0.88, 95% CI [0.24, 3.18], $p = 0.843$). This meta-analysis suggests that intra-articular BTA injection could be an efficious and safe strategy for analgesic treatment of KOA. However, evidence is limited due to the small number and heterogeneity of included studies, this urges further and stronger trials to confirm our findings.

Intraarticular botulinum toxin type A versus corticosteroid or hyaluronic acid for painful

knee osteoarthritis: A meta-analysis of head-to-head randomized controlled trials.

PMID: 38401692 | Indexed in PubMed

Intraarticular botulinum toxin type A (BTA) has been shown to be effective for painful knee osteoarthritis (KOA), while the efficacy and safety of intraarticular BTA compared to corticosteroid and hyaluronic acid (HA) remains unknown. A meta-analysis was performed to compare. A search was conducted in Medline (PubMed), CENTER (Cochrane Library), Embase (Ovid), Web of Science, Wanfang, and CNKI to find head-to-head randomized controlled trials (RCTs) directly compare the efficacy and safety between intraarticular BTA and intraarticular corticosteroid or HA for patients with painful KOA. The Cochrane Q test and estimation of I2 were used to assess heterogeneity among studies. After incorporating heterogeneity, a random-effects model was employed for data pooling. Overall, six RCTs involving 348 adults with KOA were included. Intraarticular BTA showed similar efficacy with corticosteroid as evidenced by the changes of pain visual analog scale (VAS: -0.35 [-0.97, 0.28]), total Western Ontario McMaster Universities Arthritis Index (WOMAC: 0.28 [-4.13, 4.69]), and WOMAC for pain (0.64 [-0.42, 1.70]), stiffness (-0.02 [-0.54, 0.50]), and function (0.00 [-2.99, 3.00]). Intraarticular BTA was shown to be more effective than HA in improving pain VAS (-1.31 [-1.97, -0.64]) and WOMAC for pain (-4.81 [-8.73, -0.89]), while the influence on WOMAC for knee stiffness (-1.01 [-4.43, 2.41]) and knee function (-1.86 [-6.71, 2.99]) were similar between groups. No serious adverse events were reported. Evidence from pilot RCTs suggests that intraarticular BTA may confer similar efficacy to corticosteroid for KOA, while BTA may be superior to HA for improving knee pain.

Intra-articular injection of Botulinum toxin A reduces neurogenic inflammation in CFA-induced arthritic rat model.

PMID: 27838288 | Indexed in PubMed

Currently, administration of Botulinum toxin Type A (BoNT/A) to treat arthritic pain has promising efficacy in clinical research. However, the mechanisms underlying anti-neurogenic inflammation mediated by BoNT/A remains unclear. The aim of this study was to demonstrate the effectiveness in macro and micro levels and to explore the causal mechanism of BoNT/A. Wistar rats (n = 60) were injected with 50ul complete Freund's adjuvant (CFA) in the left ankle joint capsule to establish a model of chronic monoarthritis. Pain behaviour (Evoked pain assessment) and infrared thermal imaging testing were performed at the macroscopic level to assess the effectiveness of analgesia and anti-inflammation. Western blotting and immunofluorescence staining were used at the microscopic level in an attempt to determine the mechanisms of anti-nociceptive or anti-inflammatory effects of BoNT/A. Additionally, hematoxylin-eosin staining was also used to visualise the cartilage and the synovial degenerative conditions of arthritis. By comparing the outcome of the evoked pain test and immunofluorescence staining, there was a significant improvement in BoNT/A compared with the normal saline (NS) injected control group. In addition, thermal variations showed that the temperature of ipsilateral ankle joint increased between 1 and 2 weeks following injection of CFA, but decreased after 3 weeks (still above the contralateral side). However, the temperature showed no difference between the BoNT/A group and NS group after treatment. The expression of IL-1β or TNF-α in the ankle synovial tissue was significantly decreased in the BoNT/A group compared to the NS group ($p < 0.05$). Based on the HE assessment, cartilage degeneration and infiltration of inflammatory cells in the BoNT/A group was alleviated compared to the NS group after treatment. In conclusion, we proposed the hypothesis that intra-articular BoNT/A administration does play an important role in anti-neurogenic inflammation. The possible mechanisms might be that BoNT/A prevents the release of nociceptive nerve peptides at the injection site and then suppresses the expression of inflammatory cytokines.

Use of botulinum toxin in musculoskeletal pain.

PMID: 24715952 | Indexed in PubMed

Chronic musculoskeletal pain is a common cause of chronic pain, which is associated with a total cost of $635 billion per year in the U.S. Emerging evidence suggests an anti-nociceptive action of botulinum toxin, independent of its muscle paralyzing action. This review provides a summary of data from both non-randomized and randomized clinical studies of botulinum toxin in back pain and various osteoarticular conditions, including osteoarthritis, tennis elbow, low back pain and hand pain. Three randomized controlled trials (RCTs) of small sizes provide evidence of short-term efficacy of a single intra-articular injection of 100 units of botulinum toxin A (BoNT/A) for the relief of pain and the improvement of both function and quality of life in patients with chronic joint pain due to arthritis. Three RCTs studied intramuscular BoNT/A for tennis elbow with one showing a significant improvement in pain relief compared with placebo, another one showing no difference from placebo, and the third finding that pain and function improvement with BoNT/A injection were similar to those obtained with surgical release. One RCT of intramuscular BoNT/A for low back pain found improvement in pain and function compared to placebo. Single RCTs using local injections of BoNT in patients with either temporomandibular joint (TMJ) pain or plantar fasciitis found superior efficacy compared to placebo. One RCT of intramuscular BoNT/B in patients with hand pain and carpal tunnel syndrome found improvement in pain in both BoNT/B and placebo groups, but no significant difference between groups. Most evidence is based on small studies, but the use of BoNT is supported by a single, and sometimes up to three, RCTs for several chronic musculoskeletal pain conditions. This indicates that botulinum toxin may be a promising potential new treatment for chronic refractory musculoskeletal pain. Well-designed large clinical trials are needed.

Botulinum toxin for shoulder pain.

PMID: 20824874 | Indexed in PubMed

Recent evidence suggests an anti-nociceptive effect of botulinum toxin. To compare the efficacy and safety of botulinum toxin in comparison to placebo or other treatment options for shoulder pain. We searched the Cochrane Central Register of Controlled Trials (CENTRAL) (The Cochrane Library), Ovid MEDLINE, CINAHL (via EBSCOhost), Ovid SPORTDiscus, EMBASE and Science Citation Index. Randomized controlled trials (RCTs) comparing botulinum toxin with placebo or active treatment in people with shoulder pain were included. For continuous measures we calculated mean difference (MD), and for categorical measures risk ratio (RR) (with 95% confidence interval (CI)). Six RCTs with 164 patients were included. Five RCTs in participants with post-stroke shoulder pain indicated that compared with placebo, a single intramuscular injection of botulinum toxin A significantly reduced pain at three to six months post-injection (MD -1.2 points, 95% CI -2.4 to -0.07; 0 to 10 point scale) but not at one month (MD -1.1 points, 95% CI -2.9 to 0.7). Shoulder external rotation was increased at one month (MD 9.8 degrees , 95% CI 0.2 to 19.4) but not at three to six months. Shoulder abduction, external rotation or spasticity did not differ between groups, nor did the number of adverse events (RR 1.46, 95% CI 0.6 to 24.3).One RCT in arthritis-related shoulder pain indicated that botulinum toxin reduced pain severity (MD -2.0, 95% CI -3.7 to -0.3; 10 point scale) and shoulder disability with a reduction in Shoulder Pain and Disability Index score

(MD -13.4, 95% CI -24.9 to -1.9; 100 point scale) when compared with placebo. Shoulder abduction was improved (MD 13.8 degrees, 95% CI 3.2 to 44.0). Serious adverse events did not differ between groups (RR 0.35, 95% CI: 0.11, 1.12). The results should be interpreted with caution due to few studies with small sample sizes and high risk of bias. Botulinum toxin A injections seem to reduce pain severity and improve shoulder function and range of motion when compared with placebo in patients with shoulder pain due to spastic hemiplegia or arthritis. It is unclear if the benefit of pain relief in post-stroke shoulder pain at three to six months but not at one month is due to limitations of the evidence, which includes small sample sizes with imprecise estimates, or a delayed onset of action. More studies with safety data are needed.

The efficacy and safety of Botulinum Toxin Type A in painful knee osteoarthritis: a systematic review and meta-analysis.

PMID: 31884849 | Indexed in PubMed

A systematic review and meta-analysis was carried out to evaluate the efficacy and safety of Botulinum Toxin Type A in painful knee osteoarthritis. The EMBASE and MEDLINE databases were searched to identify randomized controlled trials (RCTs) of Botulinum Toxin Type A in the treatment of painful knee osteoarthritis. The references of included literature were also searched. Five articles involving 5 RCTs including 314 patients were included in this analysis. There was a significant difference between Botulinum Toxin Type A and placebo in the visual analog scale (VAS) pain scale and Western Ontario & McMaster Universities Osteoarthritis Index (WOMAC) questionnaire score in both the short-term (≤4 weeks) and long-term (≥8 weeks) treatment period. There were no serious adverse events in the Botulinum Toxin Type A groups. This meta-analysis suggests that Botulinum Toxin Type A is effective and safe in the painful knee OA treatment. However, high-quality randomized controlled studies are still needed to further confirm our findings.

Intra-articular botulinum toxin type A for treatment of knee osteoarthritis: Clinical trial.

PMID: 30995453 | Indexed in PubMed

In recent years, there is a growing interest in new medical applications of botulinum toxin, including pain control, osteoarthritis treatment, and wound healing. While clinical applications of botulinum toxin seem promising, existing evidence regarding the therapeutic effects is still inadequate. The aim was to assess the efficacy of a single injection of abobotulinumtoxin A into the knee joint cavity to reduce pain in elderly people. We carried out a single group clinical trial in a University Hospital. Thirty participants (24 women) more than 50 years of age with knee osteoarthritis were included. Diagnosis of osteoarthritis was based on clinical and radiologic findings. We gave a single injection containing 250 units of Dysport (= 100 units of botulinum neurotoxin type A) diluted with 5 ml of normal saline. The primary outcome measure was knee pain. The secondary outcome was the patients' opinion about their knee and associated problems measured with the Knee injury and Osteoarthritis Outcome Score. The outcomes were measured at the baseline and at 4 weeks after the intervention. Within-group comparisons based on the Knee injury and Osteoarthritis Outcome Scores showed favorable results for joint pain and stiffness, sports, severity of symptoms, quality of life, and daily activities (all p-values < 0.001). Also, pain intensity, joint effusion, knee clicking and locking, and flexion-extension scores showed significant

beneficial results (all p-values ≤ 0.005). We concluded that botulinum neurotoxin type A is an effective and safe initial treatment of knee osteoarthritis with clear clinical advantages. Patients' satisfaction, minimum adverse effects in addition to single-dose prescription make the toxin as a choice for the first-line therapy of osteoarthritis at least at the short-term in elderly people. The symptom relief increases the patient's compliance and willing to participate in other therapeutic programs. REGISTRATION: Iranian Registry of Clinical Trials (IRCT) website http://www.irct.ir/, a WHO Primary Register setup, with registration code: Irct ID: IRCT20180416039323N1.